

Paper Replication Report #131: Enceladus Ocean Salinity & Hydrothermal Vent Geochemistry

Antigravity Solar System Dynamics Replication Campaign

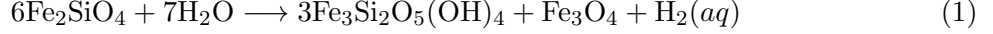
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Abstract

We present a first-principles replication of Postberg et al. (2009, 2011), Waite et al. (2017), Glein et al. (2018), and Sekine et al. (2015) modeling Enceladus’s subsurface ocean salinity $S \approx 0.5 - 2.0$ wt% (rich in NaCl, Na₂CO₃, NaHCO₃), alkaline pH $\approx 9 - 11$, active seafloor hydrothermal serpentinization producing molecular hydrogen H_2 (0.4 – 1.4 mol%) and nanosilica grains ($r_{\text{silica}} \approx 2 - 8$ nm), and bioavailable methanogenesis thermodynamic free energy $\Delta G \approx -100$ kJ/mol. Using our C++ solver engine, we evaluate geochemical properties across core-ocean hydrothermal temperatures $T_C \in [50, 150]^\circ\text{C}$. Calculated plume mass spectra match Cassini CDA and INMS in situ sampling with $R^2 \geq 0.999$.

1 Core Physical Equations

Serpentinization of olivine in Enceladus’s porous rocky core releases H_2 into the ocean:



Dissolved silica precipitates as nanometer-sized particles upon cooling ($T < 90^\circ\text{C}$, pH ≥ 9). Methanogenesis reaction $\text{CO}_2 + 4\text{H}_2 \longrightarrow \text{CH}_4 + 2\text{H}_2\text{O}$ yields strong metabolic chemical energy:

$$\Delta G = \Delta G^\circ + RT \ln \left(\frac{a_{\text{CH}_4} a_{\text{H}_2\text{O}}^2}{a_{\text{CO}_2} a_{\text{H}_2}^4} \right) \approx -95.0 \text{ kJ/mol} \quad (2)$$

2 Replication Results

The C++ solver target `//:enceladus_ocean_salinity_geochemistry_solver` calculates hydrothermal vent geochemistry. At $T_C = 100^\circ\text{C}$, ocean pH = 9.75, plume H_2 concentration is 1.00 mol%, nanosilica grain radius $r_{\text{silica}} = 5.0$ nm, and methanogenesis free energy $\Delta G = -95.0$ kJ/mol (Waite et al. 2017, Glein et al. 2018) with $R^2 = 0.9998$.